CPReady Researcher Reference: Kinematic Model Rationale, Parameter Sensitivity & Error Bounds

Document Version: 1.0 (Model v5 — Fine-Tuned 1-MPU Ground-Truth Baseline) | **Target Hardware:** ESP32 + MPU6050 (100 Hz Sampling, 100 kHz I2C) | **Ground-Truth Benchmark:** Laerdal Little Anne QC (9 Complete Sessions, 667 Compressions, 41,521 Raw Samples)

1. Executive Summary & Model Selection Rationale

During field validation of the CPReady system against the **Laerdal Little Anne QC** built-in optical/spring sensors, five kinematic depth models and two sensor topologies (`1-MPU Chest Only` vs `2-MPU Chest-minus-Base Differential`) were evaluated across all 9 complete ground-truth trials (`DATA/organized_trials/`).

Model / Sensor Configuration	Depth MAE	Depth RMSE	Mean Bias	Min Err	Max Err	Primary Physical / Mathematical Limitation
1. Historical Firmware (`packets.csv` as-run)	30.84 mm	34.12 mm	+3.40 mm	0.70 mm	46.00 mm	2-MPU common-mode cancellation (01–07) + stale `K=48.20` clamping (08–11).
2. 2-MPU Differential Double-Integration (Optimal *K*)	14.24 mm	18.79 mm	-12.93 mm	0.10 mm	29.35 mm	Base MPU shares 70%–85% of chest acceleration in-phase; coupling shifts per trial.
3. 1-MPU Legacy Peak-to-Trough ($K \cdot \Delta a \cdot T^2$)	3.83 mm	4.96 mm	+0.28 mm	0.10 mm	8.87 mm	Sharp turnarounds spike $a_{max} - a_{min}$ without increasing true depth (+8.9 mm).
4. 1-MPU Double-Integration (Uncapped Window, *K=11.20*)	3.42 mm	4.25 mm	+3.40 mm	0.10 mm	7.24 mm	Top-of-chest idle dwell (250–450 ms) integrates slow hand shifts over T^2 (+7.2 mm).
5. 1-MPU Double-Integration (Capped `MAX_PRE=18` , *K=11.20*)	2.44 mm	3.23 mm	+1.46 mm	0.10 mm	6.74 mm	Eliminates dwell inflation; +1.46 mm positive bias from external vest/foam compliance.
6. 1-MPU Double-Integration (Fine-Tuned Model v5: `MAX_PRE=22` , *K=10.85*)	**1.88 mm**	**2.49 mm**	**+0.54 mm**	**0.21 mm**	**5.43 mm**	**Selected Production Model: Global minimum Depth MAE (1.88 mm) & Stroke MAE (0.78).**

2. Mathematical Formulation of Selected Production Model (Model v5)

1. **3D Gravity Axis Projection (`GravityAxis.h`):** During the 2.0 s stationary baseline calibration (200 samples), the unit gravity vector $\hat{\mathbf{u}}$ and static magnitude g_0 are learned. Downward acceleration (in *g*) is projected onto $\hat{\mathbf{u}}$:

$$a_{net}(t_i) = -[\, a_{chest}(t_i) \cdot \hat{u} - g_0 \,]$$

2. **Bounded Pre-Onset Closed-Cycle Windowing (`CompressionTracker.h`):** To capture the pre-trigger downstroke lead-in (caused by the ~100 ms IIR filter delay, α = 0.25) while excluding inter-stroke top-of-chest idle dwell, the integration start index is bounded by `MAX_PRE_ONSET_INTEGRATION_SAMPLES` (N_{pre} = 22 samples = 220 ms):

$$i_{start} = \max(\, i_{lastStrokeEnd},\ i_{onset} - 22 \,), \qquad i_{end} = i_{recoilFinish} + 1$$

3. **Zero-Mean Closed-Cycle Double Integration (`DepthEstimator.h`):** Subtracting mean acceleration $\bar{a}$ cancels constant DC tilt leakage, and subtracting mean velocity $\bar{v}$ enforces closed-cycle boundary closure without requiring the window edges to be at rest:

$$v_k = \Sigma\,(a_{net,i} - \bar{a})\,\Delta t_i, \quad x_k = \Sigma\,(v_i - \bar{v})\,\Delta t_i, \quad d_{reported,cm} = 980.665 \cdot (\max x_k - \min x_k) \cdot (K_{depth} / 11.20)$$

where K_{depth} = `10.85` (`0.96875x` compliance factor correcting ~1.7 mm of external vest/pad compression).

3. Empirical Per-Session Verification (K = 10.85, MAX_PRE = 22)

Trial Folder	Preset	GT Strokes	Model Strokes	Stroke Err	GT Depth	Model Depth	Depth Err	GT Rate	Model Rate	Rate Err
`01_2MPU_Balanced1_a4dc7413`	BALANCED	52	53	+1	46.0 mm	51.43 mm	+5.43 mm	103	99.09	-3.91
`04_2MPU_Strict1_7c5ea2c3`	STRICT	103	104	+1	54.0 mm	54.44 mm	+0.44 mm	104	102.59	-1.41
`05_2MPU_Strict2_f501e7c8`	STRICT	69	69	0	57.0 mm	57.72 mm	+0.72 mm	109	108.53	-0.47
`06_2MPU_High1_343540cf`	HIGH	82	83	+1	60.0 mm	59.35 mm	-0.65 mm	105	104.43	-0.57
`07_2MPU_High2_e5c4ad3a`	HIGH	69	70	+1	59.0 mm	57.00 mm	-2.00 mm	101	100.71	-0.29
`08_1MPU_Strict1_6ee8938f`	STRICT	71	70	-1	54.0 mm	57.05 mm	+3.05 mm	107	104.76	-2.24
`09_1MPU_Strict2_848e796c`	STRICT	71	71	0	57.0 mm	53.82 mm	-3.18 mm	103	101.79	-1.21
`10_1MPU_Balanced1_48011a51`	BALANCED	70	70	0	58.0 mm	57.79 mm	-0.21 mm	102	100.52	-1.48
`11_1MPU_High2_44c2e739`	HIGH	80	82	+2	51.0 mm	52.26 mm	+1.26 mm	102	99.20	-2.80
Aggregate Summary (All 9 Trials)	**All**	**667**	**672**	**Min 0 | MAE 0.78 | Max 2**	**55.1 mm**	**55.65 mm**	**Min 0.21 | MAE 1.88 | RMSE 2.49 | Max 5.43 mm**	**104.0**	**102.40**	**Min 0.29 | MAE 1.60 | Max 3.91**

4. Empirical Parameter Sensitivity & Min/Max Error Bounds

4.1 Depth Calibration Scalar (`DEPTH_CALIBRATION_K` , Command: `SET_K:<val>`)

Linear sensitivity is $\partial d/\partial K = d_{phys} / 11.20 \approx 5.0$ mm per 1.0 unit of K (or 1.0 mm per 0.20 change in K).

DEPTH_CALIBRATION_K	Scale Ratio (K / 11.20)	Min Depth Error	Mean Absolute Error (MAE)	Depth RMSE	Mean Signed Bias	Max Depth Error	Signed Error Range
9.50	0.848x	0.63 mm	6.27 mm	6.82 mm	-6.27 mm	9.87 mm	[-9.87, -0.63 mm]
10.00	0.893x	1.42 mm	4.10 mm	4.54 mm	-3.74 mm	7.39 mm	[-7.39, +1.66 mm]
10.50	0.938x	0.27 mm	2.35 mm	2.77 mm	-1.22 mm	4.91 mm	[-4.91, +3.89 mm]
10.85 (Default)	**0.969x**	**0.21 mm**	**1.88 mm**	**2.49 mm**	**+0.54 mm**	**5.43 mm**	**[-3.18, +5.43 mm]**
11.00	0.982x	0.17 mm	2.10 mm	2.73 mm	+1.29 mm	6.09 mm	[-2.43, +6.09 mm]
11.20 (Unscaled)	1.000x	0.16 mm	2.66 mm	3.30 mm	+2.30 mm	6.97 mm	[-1.44, +6.97 mm]
11.50	1.027x	0.05 mm	3.81 mm	4.46 mm	+3.81 mm	8.29 mm	[+0.05, +8.29 mm]
12.00	1.071x	2.53 mm	6.32 mm	6.71 mm	+6.32 mm	10.48 mm	[+2.53, +10.48 mm]

4.2 Pre-Onset Integration Lead-In Cap (`MAX_PRE_ONSET_INTEGRATION_SAMPLES`)

MAX_PRE_ONSET_SAMPLES	Lead-In Window	Min Depth Error	Mean Absolute Error (MAE)	Depth RMSE	Mean Signed Bias	Max Depth Error	Signed Error Range
8 samples	80 ms (Clips push onset)	0.31 mm	4.14 mm	5.47 mm	-2.65 mm	11.59 mm	[-11.59, +4.00 mm]
12 samples	120 ms	0.15 mm	3.27 mm	4.17 mm	-1.65 mm	8.84 mm	[-8.84, +4.34 mm]
16 samples	160 ms	0.60 mm	2.56 mm	3.15 mm	-0.76 mm	6.42 mm	[-6.42, +4.60 mm]
18 samples	180 ms	0.45 mm	2.27 mm	2.83 mm	-0.25 mm	5.27 mm	[-5.27, +5.07 mm]
20 samples	200 ms	0.21 mm	1.94 mm	2.61 mm	+0.19 mm	5.34 mm	[-4.17, +5.34 mm]
22 samples (Default)	**220 ms (Optimal)**	**0.21 mm**	**1.88 mm**	**2.49 mm**	**+0.54 mm**	**5.43 mm**	**[-3.18, +5.43 mm]**
26 samples	260 ms	0.21 mm	2.04 mm	2.55 mm	+1.11 mm	5.47 mm	[-2.00, +5.47 mm]
Uncapped	Full `lastStrokeEnd`	0.10 mm	2.26 mm	2.91 mm	+1.77 mm	5.24 mm	[-1.29, +5.24 mm]

4.3 Recoil Arming Fraction (`RECOIL_ARM_FRACTION` , Command: `SET_ARM:<val>`) & Downstroke Trigger (`SET_TRIG:<val>`)

Parameter & Value	Preset Mapping	Stroke Min Err	Stroke MAE	Stroke Max Err	Signed Stroke Range	Depth Min Err	Depth MAE	Depth RMSE	Depth Max Err
`SET_ARM:0.08`	Too Sensitive	0	2.44 strokes	8 strokes	[0, +8]	0.13 mm	2.75 mm	3.50 mm	7.20 mm
`SET_ARM:0.10`	Legacy HIGH	0	2.00 strokes	6 strokes	[0, +6]	0.07 mm	2.41 mm	3.06 mm	5.73 mm
`SET_ARM:0.15`	BALANCED Default	0	1.00 strokes	3 strokes	[0, +3]	0.07 mm	1.95 mm	2.59 mm	5.43 mm
`SET_ARM:0.20`	**HIGH Default**	**0**	**0.67 strokes**	**2 strokes**	**[0, +2]**	**0.21 mm**	**1.83 mm**	**2.42 mm**	**5.43 mm**
`SET_ARM:0.25`	STRICT Default	0	0.67 strokes	2 strokes	[-1, +2]	0.34 mm	1.99 mm	2.69 mm	6.24 mm
`SET_TRIG:0.06`	HIGH Default	0	1.00 strokes	3 strokes	[-1, +3]	0.20 mm	1.89 mm	2.49 mm	5.43 mm
`SET_TRIG:0.08`	BALANCED Default	0	1.00 strokes	3 strokes	[-1, +3]	0.21 mm	1.89 mm	2.49 mm	5.43 mm
`SET_TRIG:0.12`	STRICT Default	0	0.78 strokes	2 strokes	[-1, +2]	0.22 mm	1.88 mm	2.48 mm	5.42 mm

5. Theoretical Perturbation Bounds (Clock Jitter, Idle Dwell, Tilt Drift & Tremor)

Perturbation Source & Range	Min Error	Mean Absolute Error (MAE)	Max Error (Worst Case)	Physical / Mathematical Bound Explanation
Sampling Jitter ±5% (9.5–10.5 ms, True-Δt Riemann)	0.00 mm	0.60 mm	1.88 mm	True microsecond `dt` integration reduces jitter error by 35% vs fixed- `dt` .
Sampling Jitter ±20% (8.0–12.0 ms, True-Δt Riemann)	0.00 mm	0.74 mm	2.78 mm	Uncompensated fixed- `dt` reaches 6.34 mm MaxErr at ±20% jitter.
Top-of-Chest Idle Dwell (400–500 ms with 0.035g hand shift)	3.29 mm	3.51 mm	4.03 mm	`MAX_PRE=22` suppresses uncapped +7.58 to +9.99 mm inflation by up to 67%.
Static DC Bias / Gravity Tilt Drift (0.00g to +0.25g)	0.00 mm	0.00 mm	0.00 mm	Exact algebraic cancellation via $\bar{a}$ subtraction over closed cycle.
Post-Pause Hand Tremor (20 mg to 110 mg on first stroke)	0.10 mm	1.18 mm	3.80 mm	`REST_LIMIT_G = 0.15g` recovers 100% of post-pause strokes up to 110 mg tremor.

6. Practical Field Tuning Decision Matrix for Researchers

Observed Field Symptom	Underlying Physical Cause	Recommended Action / Parameter Adjustment
Depth is consistently *X* mm higher/lower than a new manikin	Different spring/foam compliance on the new manikin	Run **15 s Manikin Calibration** or send `SET_K:<val>` (±0.20 in *K* = ±1.0 mm depth).
Extra 2–5 strokes counted at ends of 30-stroke bursts	Post-recoil manikin spring oscillation arming recoil	Use `BALANCED` preset or raise Recoil Arm via `SET_ARM:0.20` – `0.25` .
Very light/damped compressions (<35 mm) are missed	Peak downstroke accel stays near trigger floor	Select `HIGH` preset (`SET_SENS:HIGH` : Trigger 0.06g, Arm 0.20, MinDepth 0.25 cm).
Slow compressions (<90 CPM) with top pause over-read depth	Pre-onset window includes part of top-of-chest dwell	Lower `MAX_PRE_ONSET_INTEGRATION_SAMPLES` in `CPReadyConfig.h` from `22` to `18` .